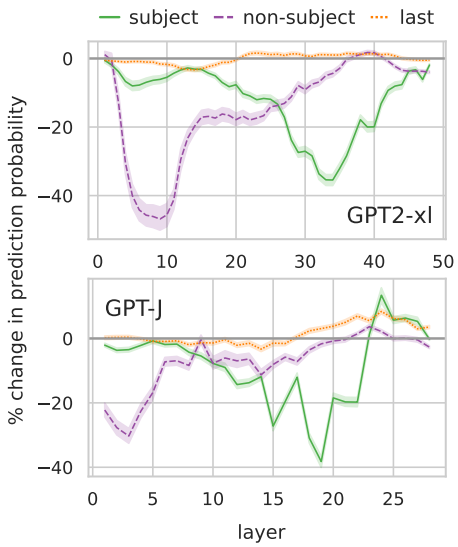


Figure 9: Relative change in the prediction probability when intervening on attention edges to the last position, for 9 layers in GPT-2 and 5 in GPT-J. Here, we do not block attention edges to the first position in the input.

put probability by ∼20% when the first position is blocked (Fig. 2) compared to <5% when it is not.

A.3 Information Flow from Subject Positions

We conjecture that, due to auto-regressivity, subject representations with critical information for the prediction are formed at the last-subject position. To verify that, we refine our interventions in §5, and block the attention edges to the last position from all the subject positions except one. We then measure the effect of these interventions on the prediction probability, which indicates from which position critical information propagates to the prediction.

Fig. 10 depicts the results for GPT-2, showing that indeed the prediction is typically damaged by $50\% - 100\%$ when the last subject-position is blocked (i.e. "first" and "before-last"), and usually remains intact when this position is not blocked (i.e. "last").

A.4 Window Size

We examine the effect of the window size hyperparameter on our information flow analysis, as conducted in §5 for the last position in GPT-2 and GPT-J. Results of this analysis with varying window sizes of $k = 1, 5, 9, 13, 17, 21$ are provided in Fig. 11 for GPT-2 and Fig. 12 for GPT-J. Overall, the same observations are consistent across different window sizes, with two prominent sites

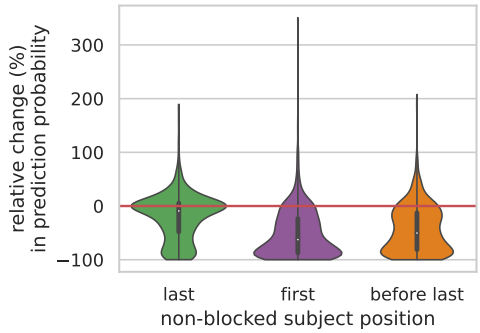


Figure 10: Relative change in the prediction probability when intervening on attention edges to the last position from different subject positions in GPT-2.

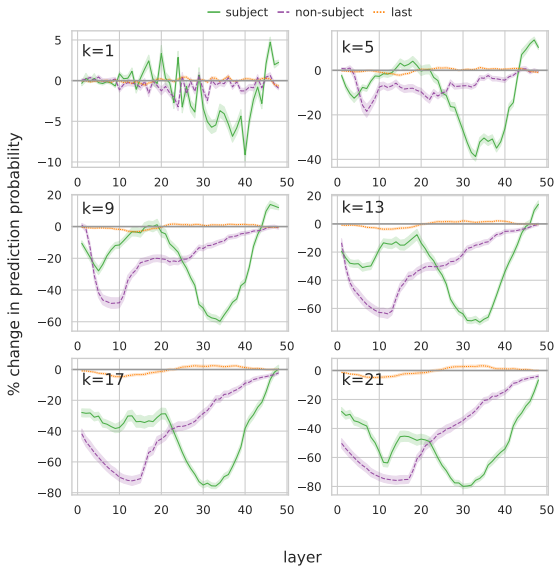


Figure 11: Relative change in the prediction probability when intervening on attention edges from different positions to the last position, for windows of k layers in GPT-2. The plots show the analysis for varying values of k.

of critical information flow to the last position – one from the relation positions in the early layers, followed by another from the subject positions in the upper layers. An exception, however, can be observed when knocking out edges from the relation positions using just a single-layer window in GPT-2; in this case, no significant change in the prediction probability is apparent. This might imply that critical information from the relation positions is processed in multiple layers. Moreover, the decrease in the prediction probability becomes more prominent when for larger values of k. This is expected, as knocking out more attention edges in the computation prevents the model from contextualizing the input properly.

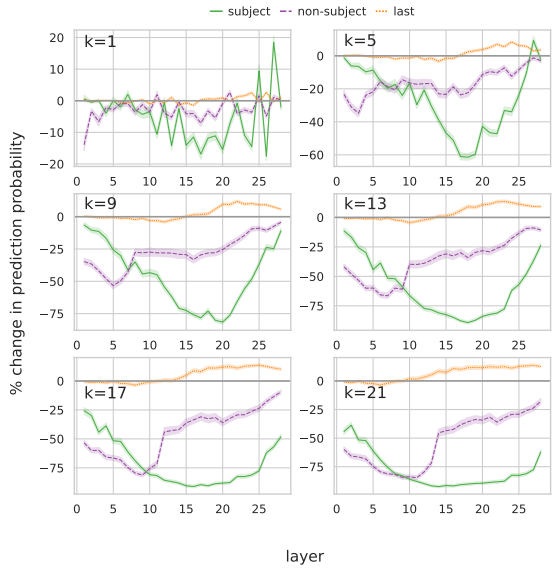


Figure 12: Relative change in the prediction probability when intervening on attention edges from different positions to the last position, for windows of k layers in GPT-J. The plots show the analysis for varying values of k.

B Gradient-based Analysis

Gradient-based feature attribution methods, also known as saliency methods, are a way to inspect what happens inside neural models for predictions of specific examples. Typically, they result in a heatmap over the input (sub)tokens, i.e., highlighted words. We distinguish between sensitivity and saliency (Ancona et al., 2019; Bastings and Filippova, 2020): methods such as Gradient-L2 (Li et al., 2016) show to which inputs the model is *sensitive*, i.e., where a small change in the input would make a large change in the output, but a method such as Gradient-times-Input (Denil et al., 2014) reflects *salience*: it shows (approximately) how much each input contributes to this particular logit value. The latter is computed as:

$$\nabla_{\mathbf{x}_i^\ell} f_c^\ell(\mathbf{x}_1^\ell, \mathbf{x}_2^\ell, \ldots, \mathbf{x}_N^\ell) \odot \mathbf{x}_i^\ell \tag{9}$$

where $\mathbf{x}_i^\ell$ is the input at position i in layer ℓ (with $\ell = 0$ being the embeddings which are used for Gradient-times-Input), and $f_c^\ell(\cdot)$ the function that takes the input sequence at layer ℓ and returns the logit for the target/predicted token c. To obtain a scalar importance score for each token, we take the L_2 norm of each vector and normalize the scores to sum to one. For our analysis we use a generalization of this method[3], and apply it not just to the input (sub)word embeddings ($\ell = 0$), but also to each intermediate transformer layer ($\ell = 1, \ldots, N$).[4]

Fig. 13 shows the per-layer gradient-times-input analysis for the input "Beats Music is owned by" for GPT-2 and GPT-J. This analysis supports our earlier findings (§6, Fig. 6) that shows the "readiness" of subject vs. non-subject positions: We observe both for both models that the subject positions remain relevant until deep into the computation, while the subject is being enriched with associations. Moreover, the input for 'owned' is relevant for the prediction in the first few layers, after which the plot suggests it is incorporated into the final position. That final position becomes more relevant for the prediction the deeper we get into the network, as it seemingly incorporates information from other positions, before it is used as the point to make the final prediction for target 'Apple' at layer 48 (28). At that point, the final position has virtually all relevance, in line with what we would expect for an auto-regressive model.

In Fig. 14 we go beyond a single example, and show per-layer gradient-times-input aggregated over the dataset (a subset of COUNTERFACT, cf. §3) for each model. We can see that the observations we made for the single example in Fig. 13 also hold in general: the subject tokens remain relevant until about $\frac{2}{3}$ into the network depth, and the relation (indicated by "further tokens") is relevant in the first few layers, after which it becomes less relevant to the prediction.

C Attributes Rate Evaluation

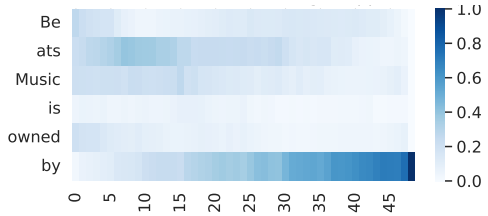
C.1 Evaluation Details

We provide details on the construction process of candidate sets for attributes rate evaluation (§6.1). Given a subject s, first we use the BM25 algorithm (Robertson et al., 1995) to retrieve 100 paragraphs from the English Wikipedia[5] with s being the query. From the resulting set, we keep only paragraphs for

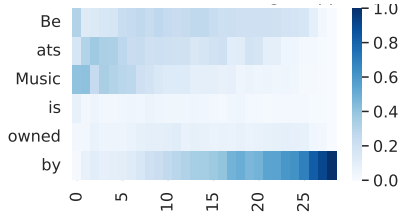
[3]Concurrently, the use of per-layer gradient-times-input, also called gradient-times-activation, was also proposed by Sarti et al. (2023) for computing *contrastive* (Yin and Neubig, 2022) per-layer explanations.

[4]We add an `Identity` layer to the output of each transformer layer block, and capture the output and gradient there, after the attention and MLP sublayers have been added to the residual. This is important especially for GPT-J, where the attention and MLP sublayers are computed in parallel, so capturing the output and gradient only at the MLP would result in incorrect attributions.

[5]We use the dump of October 13, 2021.

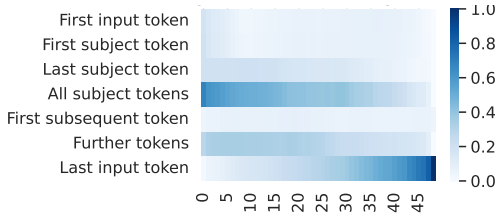


(a) GPT-2

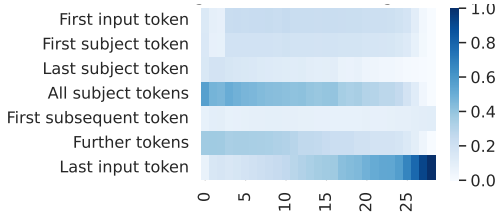


(b) GPT-J

Figure 13: Gradient-times-Activation analysis for the example "Beats Music is owned by Apple".



(a) GPT-2



(b) GPT-J

Figure 14: Gradient-times-Activation averaged over all examples.

which the subject appears as-is in their content or in the title of the page/section they are in. This is to avoid noisy paragraph that could be obtained from partial overlap with the subject (e.g. retrieving a paragraph about "2004 Summer Olympics" for the subject "2020 Summer Olympics"). This process results in 58.1 and 54.3 paragraphs per subject on average for the data subsets of GPT-2 and GPT-J, respectively.

Next, for each model, we tokenize the sets of paragraphs, remove duplicate tokens, and tokens with less than 3 characters (excluding spaces). The later is done to avoid tokens representing frequent short sub-words like `S` and `'s`. For stopwords removal, we use the list from the NLTK package.[6] This yields the final sets $\mathcal{A}_s$, of 1154.4 and 1073.1 candidate tokens on average for GPT-2 and GPT-J, respectively.

C.2 Additional Sublayer Knockout Results

We extend our analysis in §6.3, where we analyzed the contribution of the MLP and MHSA sublayers to the subject enrichment process. Specifically, we now measure the effect of knocking out these sublayers on the attribute rate at *any* successive layer, rather than on a single upper layer. Results for GPT-2 are presented in Fig. 15 (MLP sublayers knockouts) and Fig. 16 (MHSA sublayers knockouts), showing similar trends where canceling updates from the MLP sublayers decreases the attributes rate dramatically, while a more benign effect is observed when canceling MHSA updates.

[6] https://www.nltk.org/

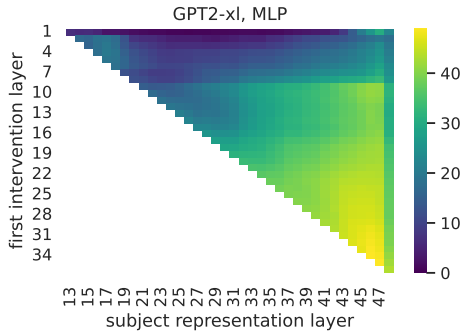


Figure 15: The attributes rate of the subject representation at different layers, when intervening on (canceling) different 10 consecutive MLP sublayers in GPT-2.

D Projection of Subject Representations

We provide additional examples for top-scoring tokens in the projection of subject representations across layers, in GPT-2 and GPT-J. Tab. 3 (Tab. 4) shows the tokens for the subject "Mark Messier" ("iPod Classic") across layers in GPT-2 and GPT-J, excluding stopwords and sub-word tokens.

E Additional Results for GPT-J

We supply here additional results for GPT-J. Fig. 17 shows the effect of canceling updates from the MHSA and MLP sublayers on the attributes rate of the subject representation at layer 22 (§6.3). Fig. 18 and Tab. 5 provide the extraction rate by the MHSA and MLP across layers and per-example extraction statistics, respectively (§7). Last, Fig. 19